

Computing the quasi-local spin of a distorted horizon: a detailed algorithm based on Korzyński’s conformal decomposition

Notes for the `KorzyńskiSpin.jl` package

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Abstract

Korzyński [1] defines the quasi-local angular momentum of a non-axisymmetric marginally outer trapped surface (MOTS) through the conformal decomposition of its two-metric and the action of the Möbius group of the sphere. These notes turn that definition into a concrete numerical algorithm: starting from the shape function $h(\theta, \varphi)$ of an apparent horizon and the Cauchy data (γ_{ij}, K_{ij}) at its collocation points, we construct the induced two-metric, the rotation one-form, uniformize the metric by two-dimensional Ricci flow, obtain the Möbius generators directly in the original coordinates from the first eigenfunctions of the round Laplacian, and evaluate the invariants $\vec{J}$ and $\vec{K}$ (eqs. (12) and (13) of [1]), the spin J , the spin axis, and the axial vector field ϕ . We work out every formula needed for a pseudospectral implementation based on spin-weighted spherical harmonics, and we collect the corrections and simplifications relative to the algorithm skeleton in `GOAL.md`; the most important one is that no explicit “good” coordinate system needs to be constructed and the Ricci flow never needs to be “undone”.

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1 Notation and conventions

- Spacetime signature $(-+++)$; geometric units $c = 1$; we keep Newton's constant G explicit (set $G = 1$ in code).
- Greek indices $\mu, \nu, \dots$ are four-dimensional, lower-case Latin indices $i, j, k, \dots$ are spatial (three-dimensional), capital Latin indices $A, B, C, \dots$ are two-dimensional indices on the horizon surface $\Delta \simeq S^2$, with coordinates (θ, φ) .
- γ_{ij} is the spatial metric of the Cauchy slice Σ , n^μ its future-pointing unit normal, and the extrinsic curvature uses the ADM/numerical-relativity sign convention

$$K_{ij} = -\frac{1}{2}\mathcal{L}_n\gamma_{ij} = -\gamma_i^\mu\gamma_j^\nu\nabla_\mu n_\nu. \quad (1)$$

With the opposite (Wald) convention all components of $\vec{J}$ and $\vec{K}$, and the axial vector ϕ , flip sign, while $J = |\vec{J}|$ is unaffected.

- q_{AB} is the induced (positive definite) metric on Δ , D_A its Levi-Civita connection, Δ_q its Laplace–Beltrami operator, ϵ_{AB} its area two-form, and $R[q]$ its Ricci scalar ($R = 2K_{\text{Gauss}}$; for the unit round sphere $R = 2$).
- The chart (θ, φ) is assumed right-handed as seen from outside the surface, and we orient ϵ_{AB} such that $\epsilon_{\theta\varphi} = +\sqrt{\det q} > 0$. Reversing the orientation flips $\vec{J}$, $\vec{K}$ and ϕ .
- A ring denotes quantities of the *unit-round representative* $\hat{q}_{AB}$ produced by the Ricci flow (area 4π , $R[\hat{q}] = 2$): $\hat{\epsilon}_{AB}$, $\hat{D}_A$, $\hat{\Delta}$. A hat denotes quantities of the *unit coordinate sphere* $\hat{q} = d\theta^2 + \sin^2\theta d\varphi^2$ in the given chart, which serves only as a pseudospectral bookkeeping device (Section 4); $\hat{q}$ is *not* in general conformal to q in this chart.
- In two dimensions we use repeatedly, for $\tilde{q} = e^{2u}q$,

$$\tilde{\epsilon}_{AB} = e^{2u}\epsilon_{AB}, \quad \Delta_{\tilde{q}}f = e^{-2u}\Delta_q f, \quad R[\tilde{q}] = e^{-2u}(R[q] - 2\Delta_q u), \quad (2)$$

the identities $\epsilon_A^C \epsilon_C^B = -\delta_A^B$, $\epsilon^{AC} \epsilon_{BC} = \delta_B^A$, and the conformal invariance of the Hodge star on one-forms, $(\star\alpha)_A = \epsilon_A^B \alpha_B$, which is unchanged under $q \rightarrow e^{2u}q$.

2 Korzyński's definition of angular momentum

This section summarizes the parts of [1] that the algorithm implements; equation numbers in parentheses refer to that paper.

2.1 MOTS geometry and the rotation one-form

A MOTS Δ carries the metric q_{AB} , the area form ϵ , and the *rotation one-form*

$$\omega_A = -(\nabla_A l^\mu) k_\mu = (\nabla_A k^\mu) l_\mu, \quad l^\mu k_\mu = -1, \quad (3)$$

where l^μ, k^μ are the outgoing and ingoing null normals (paper eq. (1)). Under a rescaling $l \rightarrow C l$, $k \rightarrow C^{-1} k$ (equivalently: a change of the slicing through Δ) the rotation form changes by a gradient,

$$\omega_A \rightarrow \omega_A + \partial_A \ln C. \quad (4)$$

Every standard definition of quasi-local angular momentum has the form (paper eq. (2))

$$J_\phi = -\frac{1}{8\pi G} \int_\Delta \omega(\phi) \epsilon \quad (5)$$

for some *axial* vector field ϕ on Δ (two poles, closed orbits, affine parameter period 2π). The entire problem is the choice of ϕ when Δ has no symmetry.

2.2 Conformally spherical coordinates and the Möbius group

On any oriented (S^2, q) there exist global *conformally spherical coordinates* (CSCS) (θ, φ) in which

$$q = F(\theta, \varphi) (d\theta^2 + \sin^2 \theta d\varphi^2), \quad F > 0 \quad (6)$$

(uniformization; constructible by Ricci flow [2, 3]). Such coordinates are unique only up to the six-parameter Möbius group, the connected component of $SO(1, 3)$, generated by the three rotations

$$\begin{aligned} \phi_1 &= -\sin \varphi \partial_\theta - \cot \theta \cos \varphi \partial_\varphi, \\ \phi_2 &= \cos \varphi \partial_\theta - \cot \theta \sin \varphi \partial_\varphi, \\ \phi_3 &= \partial_\varphi, \end{aligned} \quad (7)$$

and the three proper conformal generators

$$\begin{aligned} \xi_1 &= -\cos \theta \cos \varphi \partial_\theta + \frac{\sin \varphi}{\sin \theta} \partial_\varphi, \\ \xi_2 &= -\cos \theta \sin \varphi \partial_\theta - \frac{\cos \varphi}{\sin \theta} \partial_\varphi, \\ \xi_3 &= \sin \theta \partial_\theta, \end{aligned} \quad (8)$$

with commutators $[\phi_i, \phi_j] = -\epsilon_{ijk} \phi_k$, $[\xi_i, \xi_j] = +\epsilon_{ijk} \phi_k$, $[\xi_i, \phi_j] = -\epsilon_{ijk} \xi_k$. Every unit combination $n_i \phi_i$ ($n_i n_i = 1$) is axial; no combination of the ξ_i is.

The finite action of the proper conformal transformation generated by $\lambda n_i \xi_i$ ($n_i n_i = 1$, $\lambda \in \mathbb{R}$) on the generators themselves is (paper eqs. (10)–(11))

$$\begin{aligned} \tilde{\phi}_i &= n_i (n_k \phi_k) + \cosh \lambda (\phi_i - n_i (n_k \phi_k)) + \sinh \lambda \epsilon_{ijk} n_j \xi_k, \\ \tilde{\xi}_i &= n_i (n_k \xi_k) + \cosh \lambda (\xi_i - n_i (n_k \xi_k)) - \sinh \lambda \epsilon_{ijk} n_j \phi_k. \end{aligned} \quad (9)$$

2.3 The invariants $\vec{J}$ and $\vec{K}$

Given a CSCS one defines the two triples (paper eqs. (12)–(13))

$$\boxed{J_i = -\frac{1}{8\pi G} \int_{\Delta} \omega(\phi_i) \epsilon, \quad K_i = -\frac{1}{8\pi G} \int_{\Delta} \omega(\xi_i) \epsilon.} \quad (10)$$

Note that ϵ is the *physical* area form of q , while ϕ_i and ξ_i are the round-sphere generators (7)–(8) of the chosen CSCS. Under the rotation subgroup $SO(3)$, $\vec{J}$ and $\vec{K}$ rotate as Euclidean 3-vectors. Under the proper conformal transformation (9), with $\vec{\beta} = \tanh \lambda \hat{n}$, $\gamma = (1 - \beta^2)^{-1/2}$, they transform exactly like the electric and magnetic fields under a Lorentz boost (paper eqs. (14)–(15)):

$$\begin{aligned} \vec{J}' &= \gamma(\vec{J} + \vec{\beta} \times \vec{K}) - \frac{\gamma^2}{\gamma + 1} \vec{\beta}(\vec{\beta} \cdot \vec{J}), \\ \vec{K}' &= \gamma(\vec{K} - \vec{\beta} \times \vec{J}) - \frac{\gamma^2}{\gamma + 1} \vec{\beta}(\vec{\beta} \cdot \vec{K}). \end{aligned} \quad (11)$$

The two $SO(1, 3)$ invariants are (paper eqs. (16)–(17))

$$A = |\vec{J}|^2 - |\vec{K}|^2, \quad B = \vec{K} \cdot \vec{J}. \quad (12)$$

2.4 Spin, axial vector, and gauge fixing

If $A^2 + B^2 > 0$ there is a proper conformal transformation making $\vec{J}$ and $\vec{K}$ parallel (or one of them zero): take

$$\vec{\beta} = \beta \frac{\vec{J} \times \vec{K}}{|\vec{J} \times \vec{K}|}, \quad \beta^2 - S\beta + 1 = 0, \quad S = \frac{|\vec{J}|^2 + |\vec{K}|^2}{|\vec{J} \times \vec{K}|}, \quad \beta = \frac{S - \sqrt{S^2 - 4}}{2} \in (0, 1) \quad (13)$$

(the root with $0 < \beta < 1$; note $S \geq 2$ always, by $|\vec{J}|^2 + |\vec{K}|^2 \geq 2|\vec{J}||\vec{K}| \geq 2|\vec{J} \times \vec{K}|$). In the resulting frame the spin and axial vector are defined as (paper Definition and eq. (18))

$$J = |\vec{J}'|, \quad \phi = \frac{J'_i}{|\vec{J}'|} \tilde{\phi}_i \quad (\text{or } \phi = \frac{K'_i}{|\vec{K}'|} \tilde{\phi}_i \text{ if } \vec{J}' = 0). \quad (14)$$

Equivalently, and more conveniently, J has the manifestly $SO(1, 3)$ -invariant form (paper eq. (21))

$$\boxed{J = \sqrt{\frac{A + \sqrt{A^2 + 4B^2}}{2}}.} \quad (15)$$

For $\vec{K} = 0$ this reduces to $J = |\vec{J}|$, as it must. If $A = B = 0$ (the “plane wave” case) then $J = 0$ and no axis exists.

Finally, because of the gauge freedom (4) the paper fixes the normalization of the null normals by requiring that in the Hodge decomposition

$$\omega = \star df + dg \quad (16)$$

the exact part vanishes, $dg = 0$, i.e. $d\star\omega = 0$ (paper eq. (22)). Since $\star$ on one-forms is conformally invariant, the decomposition is the same with respect to q and $\hat{q}$. In practice one never rescales the normals: one simply replaces ω by its co-exact part

$$\omega^{\text{inv}} = \star df = \omega - dg \quad (17)$$

in (10). This gauge fixing is what makes the result independent of the choice of Cauchy slicing through the horizon (e.g. tilted slicings of Kerr).

3 The algorithm in detail

3.1 Inputs

1. A centre c^i and a shape function $h(\theta, \varphi) > 0$, given by spherical-harmonic coefficients, such that the horizon is

$$x^i(\theta, \varphi) = c^i + h(\theta, \varphi) \hat{r}^i(\theta, \varphi), \quad \hat{r}^i = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta), \quad (18)$$

i.e. the surface is star-shaped about c^i (as produced by `ApparentHorizonFinder`).

2. The spatial metric γ_{ij} and extrinsic curvature K_{ij} evaluated at the collocation points $x^i(\theta_p, \varphi_p)$ of a spin-weighted spherical-harmonic grid.

Remark 1. *No spatial derivatives of γ_{ij} or K_{ij} are needed: all derivatives that enter the algorithm are tangential to the surface and are taken pseudospectrally on the sphere. If $\partial_k \gamma_{ij}$ is available it can be used for an independent cross-check of the spectral derivatives of q_{AB} .*

3.2 Surface geometry

The tangent vectors and induced metric are

$$e_A^i = \partial_A x^i = (\partial_A h) \hat{r}^i + h \partial_A \hat{r}^i, \quad q_{AB} = \gamma_{ij} e_A^i e_B^j. \quad (19)$$

The angular derivatives $\partial_A h$ are evaluated spectrally from the spherical-harmonic coefficients of h . The area form and the physical area are

$$\epsilon_{AB} = \sqrt{\det q} (d\theta \wedge d\varphi)_{AB}, \quad \mathcal{A} = \oint_{\Delta} \epsilon = \int_0^\pi \int_0^{2\pi} \sqrt{\det q} d\theta d\varphi. \quad (20)$$

The outward unit normal of Δ within Σ is obtained without any derivatives of γ : the covector

$$\sigma_i = [ijk] e_\theta^j e_\varphi^k \quad ([ijk] \text{ the flat permutation symbol}) \quad (21)$$

annihilates both tangents (this requires no metric), and

$$s^i = \frac{\gamma^{ij} \sigma_j}{\sqrt{\gamma^{kl} \sigma_k \sigma_l}}, \quad \text{sign fixed by } s^i (x_i - c_i) > 0. \quad (22)$$

3.3 The rotation one-form from ADM data

Let n^μ be the future unit normal of Σ and s^μ the outward unit normal of Δ in Σ , and set

$$l^\mu = \frac{1}{\sqrt{2}} (n^\mu + s^\mu), \quad k^\mu = \frac{1}{\sqrt{2}} (n^\mu - s^\mu), \quad l^\mu k_\mu = -1. \quad (23)$$

Inserting these into (3) and using $n_\mu \nabla_\nu n^\mu = s_\mu \nabla_\nu s^\mu = 0$ and $n_\mu \nabla_\nu s^\mu = -s^\mu \nabla_\nu n_\mu$ (from $n \cdot s = 0$),

$$\omega_A = -\frac{1}{2} e_A^\nu (n - s)_\mu \nabla_\nu (n + s)^\mu = e_A^\nu s^\mu \nabla_\nu n_\mu. \quad (24)$$

Both e_A and s are tangent to Σ , so with the convention (1),

$$\boxed{\omega_A = -K_{ij} e_A^i s^j.} \quad (25)$$

Consequently (5) becomes $J_\phi = \frac{1}{8\pi G} \oint K_{ij} \phi^i s^j \epsilon$ with $\phi^i = \phi^A e_A^i$, which is the familiar isolated-horizon / Brown–York integrand of [5]. Note that the constant boost factor $1/\sqrt{2}$ in (l, k) is pure gauge (4) and drops out after the Hodge projection below; (25) holds for any constant normalization.

3.4 Gauge fixing: Hodge decomposition of ω

On S^2 there are no harmonic one-forms, so (16) is the complete Hodge decomposition. Taking the divergence of (16) and using $D^A(\epsilon_A{}^B D_B f) = \epsilon^{AB} D_A D_B f = 0$,

$$\Delta_q g = D^A \omega_A = \frac{1}{\sqrt{\det q}} \partial_A (\sqrt{\det q} q^{AB} \omega_B), \quad (26)$$

a Poisson equation with zero-mean right-hand side, solved (up to an irrelevant constant) pseudospectrally, after which

$$\omega_A^{\text{inv}} = \omega_A - \partial_A g. \quad (27)$$

Because $\star$ on one-forms and the splitting exact $\oplus$ co-exact are conformally invariant, solving (26) with q or with $\hat{q}$ gives the same g — a useful consistency check. The potential f is not needed.

Remark 2. *This step cannot be skipped. The fields ϕ_i are divergence-free with respect to the round area form $\hat{\epsilon}$, but not with respect to the physical $\epsilon = F \hat{\epsilon}$; hence $\oint (\text{d}g)(\phi_i) \epsilon = - \oint g \phi_i (\ln F) \epsilon \neq 0$ in general, and without the projection the J_i would depend on the slicing through the horizon.*

3.5 Ricci scalar of the two-metric

The flow in Section 3.6 needs $R[q]$. In any chart,

$$\Gamma_{AB}^C = \frac{1}{2} q^{CD} (\partial_A q_{DB} + \partial_B q_{AD} - \partial_D q_{AB}), \quad R = q^{AB} (\partial_C \Gamma_{AB}^C - \partial_A \Gamma_{CB}^C + \Gamma_{CD}^C \Gamma_{AB}^D - \Gamma_{AD}^C \Gamma_{CB}^D). \quad (28)$$

Coordinate components of q_{AB} are singular at the poles of the chart, so in practice (28) is evaluated in the spin-weighted form of Section 4: with $\hat{\nabla}$ the connection of the unit coordinate sphere $\hat{q}$ and

$$C_{BC}^A = \frac{1}{2} q^{AD} (\hat{\nabla}_B q_{DC} + \hat{\nabla}_C q_{BD} - \hat{\nabla}_D q_{BC}) \quad (29)$$

(a globally regular tensor field),

$$R[q] = q^{AB} (\hat{q}_{AB} + \hat{\nabla}_C C_{AB}^C - \hat{\nabla}_A C_{CB}^C + C_{CD}^C C_{AB}^D - C_{AD}^C C_{CB}^D), \quad (30)$$

where $\hat{R}_{AB} = \hat{q}_{AB}$ is the Ricci tensor of the unit sphere. Similarly, for any scalar f ,

$$\Delta_q f = q^{AB} (\hat{\nabla}_A \hat{\nabla}_B f - C_{AB}^C \partial_C f). \quad (31)$$

3.6 Uniformization by normalized Ricci flow

First rescale to unit area: let

$$\bar{q}_{AB} = \frac{4\pi}{\mathcal{A}} q_{AB}, \quad \oint \bar{\epsilon} = 4\pi, \quad (32)$$

so that by Gauss–Bonnet the average curvature is $\bar{R}_{\text{avg}} = 2$. In two dimensions $R_{AB} = \frac{1}{2} R q_{AB}$, so the normalized Ricci flow

$$\partial_t q_{AB}(t) = (2 - R(t)) q_{AB}(t), \quad q(0) = \bar{q}, \quad (33)$$

preserves the conformal class and the total area: writing $q(t) = e^{2u(t)} \bar{q}$ and using (2), (33) is the single scalar parabolic equation

$$\boxed{\partial_t u = 1 - \frac{1}{2} R(t), \quad R(t) = e^{-2u} (\bar{R} - 2\bar{\Delta} u), \quad u(0) = 0,} \quad (34)$$

with $\bar{R} = R[\bar{q}]$ and $\bar{\Delta} = \Delta_{\bar{q}}$ computed once via (30)–(31). By Hamilton and Chow [2, 3] the flow exists for all t and converges exponentially, for *any* initial metric on S^2 , to the constant curvature metric: $u(t) \rightarrow u_\infty$ with

$$\dot{q}_{AB} = e^{2u_\infty} \bar{q}_{AB}, \quad R[\dot{q}] = 2, \quad \oint \dot{\epsilon} = 4\pi. \quad (35)$$

This $\dot{q}$ is the unit-round representative of the conformal class of q , expressed in the original chart. Equivalently, the conformal factor of (6) is $F = (\mathcal{A}/4\pi) e^{-2u_\infty}$.

Time integration. The linearization of (34) is $\partial_t \delta u = e^{-2u} \bar{\Delta} \delta u - R(t) \delta u$, so explicit integrators need $\Delta t \lesssim c \min(e^{2u}) / \ell_{\max}(\ell_{\max} + 1)$ (times the smallest eigenvalue scale of $\bar{q}$ relative to $\dot{q}$); an adaptive explicit Runge–Kutta scheme is adequate at moderate resolution. At high $\ell_{\max}$ use an IMEX or exponential integrator, treating the constant-coefficient operator $\Lambda = \hat{\Delta}$ (diagonal in the spherical-harmonic basis, eigenvalues $-\ell(\ell+1)$) implicitly and the remainder $1 - \frac{1}{2}R - \kappa \hat{\Delta} u$ explicitly, with a constant $\kappa \approx \max(e^{-2u} \lambda_{\bar{q}})$ chosen for stability. Monitor $\|R(t) - 2\|_\infty$ and stop when it reaches the requested tolerance; also monitor $\oint e^{2u} \dot{\epsilon} = 4\pi$ (preserved by (34) up to time-discretization error; renormalize u by a constant if it drifts).

Newton polish (recommended). The fixed point of (34) satisfies the Liouville-type elliptic equation

$$\mathcal{N}[u] \equiv \bar{\Delta} u - \frac{1}{2} \bar{R} + e^{2u} = 0, \quad (36)$$

with Fréchet derivative $\mathcal{N}'[u] \delta u = \bar{\Delta} \delta u + 2e^{2u} \delta u$. Note that this nonlinearity is essential: no substitution $v = F^s$ of the conformal factor $F = e^{2u}$ linearizes (36). E.g. $v = 1/F$ gives $\bar{\Delta} v + \bar{R} v - |\bar{\nabla} v|^2/v = 2$, which misses linearity only by the gradient term, but the term never cancels for any power (in contrast to the Yamabe problem in $n \geq 3$ dimensions, where $\tilde{q} = \psi^{4/(n-2)} q$ removes it identically); see the puncture method below for a genuinely linear alternative.

One subtlety: $\mathcal{N}'$ is *not* sign definite, and at the solution it is singular. Indeed $\mathcal{N}'[u_\infty] = e^{2u_\infty}(\hat{\Delta} + 2)$, whose kernel is exactly the $\ell = 1$ eigenspace $\text{span}\{\chi_i\}$ of Section 3.7. This must be so: the solutions of (36) form a three-parameter family (the Möbius orbit of round representatives, all equally acceptable), and the kernel directions are its tangent, $\delta u = \chi_i$, matching the conformal Killing flow $\mathcal{L}_{\xi, \dot{q}} = 2\chi_i \dot{q}$. Constants are a positive direction ($(\hat{\Delta} + 2)c = 2c$) and $\ell \geq 2$ modes are negative. In practice: use MINRES (not CG) for the Newton steps and take minimal-norm steps, or deflate/project out the three near-kernel modes; drift along the kernel is pure gauge and harmless. Convergence is then quadratic transverse to the gauge orbit. The most efficient strategy is: integrate (34) until $\|R - 2\|_\infty \lesssim 10^{-2}$ (the flow is unconditionally globally convergent and needs no good initial guess), then polish with Newton to round-off. Pure Ricci flow alone is also fine, merely slower.

A fully linear alternative: flattening through a puncture. Although (36) cannot be linearized by a change of variables, genus-zero uniformization *can* be reduced to linear solves by changing the route: flatten first, then map to the sphere by inverse stereographic projection [6]. Pick an arbitrary puncture point $p \in \Delta$ and solve the *linear* Poisson equation

$$\bar{\Delta} w = \frac{1}{2} \bar{R} - 4\pi \delta_p \quad (37)$$

(solvable on the closed surface because both terms integrate to 4π by Gauss–Bonnet; near p , $w \simeq -2 \ln r$). Then $e^{2w} \bar{q}$ is flat on $\Delta \setminus \{p\}$ and complete — it is the Euclidean plane, with p at infinity.

Its Cartesian coordinates (x, y) are a conjugate pair of $\bar{q}$ -harmonic functions on $\Delta \setminus \{p\}$ with a simple pole at p (harmonicity of functions is conformally invariant in two dimensions), obtained from a second linear solve. Inverse stereographic projection of $z = x + iy$ then gives the round representative and the eigenfunction triple in closed form,

$$\chi_1 + i\chi_2 = \frac{2z}{1 + |z|^2}, \quad \chi_3 = \frac{|z|^2 - 1}{|z|^2 + 1}, \quad \dot{q} = \frac{4 \, dz \, d\bar{z}}{(1 + |z|^2)^2}, \quad (38)$$

so this route replaces *both* the flow of Section 3.6 *and* the eigenproblem of Section 3.7. The arbitrary choices (the puncture p , the normalization of the conjugate pair) move the resulting CSCS by exactly a Möbius transformation, which the invariants of Section 3.9 are designed to absorb. The price is the singular sources: to retain spectral accuracy the logarithm and the pole must be split off analytically (solve for the smooth remainders), which on a curved background $\bar{q}$ requires local expansions around p . Since the Newton polish above costs only a handful of smooth linear solves anyway, we keep flow+Newton as the mainline and recommend the puncture method as an independent cross-check of the uniformization stage.

3.7 The first eigenfunctions of the round Laplacian

On the unit round sphere the lowest nonzero eigenvalue of the Laplace–Beltrami operator is -2 , with the three-dimensional eigenspace spanned by the $\ell = 1$ harmonics. We therefore solve

$$\dot{\Delta}\chi = -2\chi \quad \Longleftrightarrow \quad \bar{\Delta}\chi = -2e^{2u_\infty}\chi \quad (39)$$

(using $\dot{\Delta} = e^{-2u_\infty}\bar{\Delta}$) for the three eigenfunctions χ_i , $i = 1, 2, 3$ — these are the functions called x, y, z in `GOAL.md`. The right form for numerics is the *generalized symmetric eigenproblem* on the right of (39): $\bar{\Delta}$ is self-adjoint with respect to the measure $\bar{e}$, and the weight e^{2u_∞} is positive, so the eigenvalues are real and eigenfunctions belonging to different eigenvalues are orthogonal in

$$\langle a, b \rangle = \oint a b e^{2u_\infty} \bar{e} = \oint a b \dot{e}. \quad (40)$$

In exact arithmetic the eigenvalue -2 is exactly triply degenerate; numerically one finds a tight cluster of three eigenvalues near -2 separated by a gap from 0 above and from ≈ -6 ($\ell = 2$) below, so a shift-invert Krylov method targeted at -2 is robust (Section 4).

The cluster fixes only the 3-space $\text{span}\{\chi_i\}$; normalize and orient as follows:

1. **Orthonormalization.** Compute the Gram matrix $G_{ij} = \frac{3}{4\pi} \oint \chi_i \chi_j \dot{e}$ and replace $\chi_i \leftarrow (G^{-1/2})_{ij} \chi_j$ (symmetric/Löwdin orthonormalization), so that

$$\oint \chi_i \chi_j \dot{e} = \frac{4\pi}{3} \delta_{ij}, \quad \oint \chi_i \dot{e} = 0 \quad (41)$$

(the zero mean is automatic: $\chi_i \perp \text{constants}$).

2. **Handedness.** Compute

$$\mathcal{O} = \oint \chi_1 \dot{e}^{AB} \partial_A \chi_2 \partial_B \chi_3 \dot{e}; \quad (42)$$

if $\mathcal{O} < 0$, swap $\chi_1 \leftrightarrow \chi_2$ (or flip the sign of one χ_i).

By the classical rigidity of the first eigenfunctions (Takahashi [7]; the map $p \mapsto \vec{\chi}(p)$ is an isometric embedding of $(\Delta, \mathring{q})$ onto the unit sphere in $\mathbb{R}^3$), the normalized triple satisfies *pointwise*

$$\delta^{ij} \chi_i \chi_j = 1, \quad \delta^{ij} \partial_A \chi_i \partial_B \chi_j = \mathring{q}_{AB}, \quad (43)$$

and, with the correct handedness,

$$\chi_1 d\chi_2 \wedge d\chi_3 + \chi_2 d\chi_3 \wedge d\chi_1 + \chi_3 d\chi_1 \wedge d\chi_2 = \mathring{e}, \quad \text{i.e.} \quad \frac{1}{2} \epsilon^{ijk} \chi_i \mathring{e}^{AB} \partial_A \chi_j \partial_B \chi_k = 1. \quad (44)$$

Equation (44) integrated over the sphere shows that the handedness integral (42) evaluates to $\mathcal{O} = +\frac{4\pi}{3}$ for a right-handed triple (and $-\frac{4\pi}{3}$ otherwise), so the test is unambiguous. The residuals of (39), (43) and (44) are sharp diagnostics of the overall accuracy of the uniformization and eigenvalue steps.

The remaining freedom in $\{\chi_i\}$ is exactly one global rotation $\chi_i \rightarrow \Lambda_i^j \chi_j$, $\Lambda \in SO(3)$. No “north pole” or “zero meridian” need be selected: under this freedom $\vec{J}$ and $\vec{K}$ rotate covariantly, the invariants (12), the spin (15) and the axial field (14) are unchanged.

3.8 The Möbius generators in the original chart

This is the step where the present algorithm departs from (and simplifies) the skeleton in `GOAL.md`: the generators (7)–(8) can be written covariantly in terms of $\mathring{q}$ and χ_i and therefore evaluated *directly in the original coordinates*, without ever constructing the good coordinates (θ_1, φ_1) and without “undoing” the Ricci flow.

Lemma 1. *Let $\mathring{q}$ be a unit-round metric on S^2 and χ_i a normalized, right-handed triple of first eigenfunctions as in Section 3.7. Then the rotation and proper conformal generators of the Möbius group of $\mathring{q}$ are*

$$\boxed{\phi_i^A = \mathring{e}^{AB} \partial_B \chi_i, \quad \xi_i^A = -\mathring{q}^{AB} \partial_B \chi_i.} \quad (45)$$

Moreover $\phi_i^A = \epsilon_{ijk} \chi_j \mathring{q}^{AB} \partial_B \chi_k$.

Proof. Both sides of each identity are tensorial, so it suffices to verify them in coordinates (θ, φ) adapted to $\mathring{q}$, which exist by (43): $\mathring{q} = d\theta^2 + \sin^2\theta d\varphi^2$ and

$$\chi_1 = \sin\theta \cos\varphi, \quad \chi_2 = \sin\theta \sin\varphi, \quad \chi_3 = \cos\theta. \quad (46)$$

With $\mathring{e}_{\theta\varphi} = \sin\theta$, i.e. $\mathring{e}^{\theta\varphi} = 1/\sin\theta$:

$$\begin{aligned} \mathring{e}^{AB} \partial_B \chi_3 : \quad \mathring{e}^{\theta\varphi} \partial_\varphi \cos\theta &= 0, \quad \mathring{e}^{\varphi\theta} \partial_\theta \cos\theta = \left(-\frac{1}{\sin\theta}\right)(-\sin\theta) = 1 \implies \partial_\varphi = \phi_3. \\ \mathring{e}^{AB} \partial_B \chi_1 : \quad \mathring{e}^{\theta\varphi} \partial_\varphi (\sin\theta \cos\varphi) &= -\sin\varphi, \quad \mathring{e}^{\varphi\theta} \partial_\theta (\sin\theta \cos\varphi) = -\cot\theta \cos\varphi \implies \phi_1, \end{aligned}$$

and analogously for ϕ_2 , reproducing (7). For the second family, $-\mathring{q}^{AB} \partial_B \chi_3$ has components $(-\partial_\theta \cos\theta, 0) = (\sin\theta, 0)$, i.e. $\sin\theta \partial_\theta = \xi_3$, and $-\mathring{q}^{AB} \partial_B \chi_1 = -\cos\theta \cos\varphi \partial_\theta + \frac{\sin\varphi}{\sin\theta} \partial_\varphi = \xi_1$, etc., reproducing (8). The alternative formula for ϕ_i follows from $\epsilon_{ijk} \chi_j \partial_A \chi_k = \mathring{e}_{AB} \mathring{q}^{BC} \partial_C \chi_i$, which is readily checked on (46) (both sides are the rotation covector field about the i -axis). $\square$

Concretely, at every collocation point:

$$\mathring{q}_{AB} = e^{2u_\infty} \frac{4\pi}{\mathcal{A}} q_{AB}, \quad \mathring{e}^{AB} = \frac{\pm 1}{\sqrt{\det \mathring{q}}} \quad (\mathring{e}^{\theta\varphi} = +1/\sqrt{\det \mathring{q}}), \quad \phi_i^A, \xi_i^A \text{ from (45)}. \quad (47)$$

The fields (45) are normalization-correct (2π -periodic orbits for $n_i \phi_i$) precisely because $\mathring{q}$ has area 4π and the χ_i obey (41)–(43); this is why the area normalization of Section 3.6 matters.

3.9 Invariants, boost, spin, axis, and axial vector field

With ω^{inv} from Section 3.4, the generators from Section 3.8, and the *physical* area form ϵ of q :

$$J_i = -\frac{1}{8\pi G} \oint_{\Delta} \omega_A^{\text{inv}} \phi_i^A \epsilon, \quad K_i = -\frac{1}{8\pi G} \oint_{\Delta} \omega_A^{\text{inv}} \xi_i^A \epsilon. \quad (48)$$

Then:

1. **Spin.** Compute A and B from (12) and

$$J = \sqrt{\frac{A + \sqrt{A^2 + 4B^2}}{2}}. \quad (49)$$

This requires no frame change and is the primary output. If $A^2 + B^2 = 0$ (to within tolerance), report $J = 0$ and stop: no axis or axial field exists.

2. **Boost to the parallel frame.** If $|\vec{J} \times \vec{K}| > \text{tol}$, compute S , β , and $\hat{n} = (\vec{J} \times \vec{K})/|\vec{J} \times \vec{K}|$ from (13), set $\lambda = \text{artanh } \beta$, and obtain $\vec{J}'$ (and $\vec{K}'$, as a check: $\vec{J}' \times \vec{K}' \approx 0$) from (11) and the transformed rotation generators $\tilde{\phi}_i$ from (9), all evaluated pointwise in the original chart. If $\vec{J} \times \vec{K} \approx 0$ already, set $\tilde{\phi}_i = \phi_i$, $\vec{J}' = \vec{J}$. (Consistency: $|\vec{J}'|$ must equal (15) up to round-off, since in the parallel frame $A = J'^2 - K'^2$, $B = \pm J'K'$ and (15) returns J' .)
3. **Axis and axial vector field.** The spin axis (unit vector in the $\mathbb{R}^3$ of the χ'_i , i.e. relative to the boosted CSCS frame, defined up to the global $SO(3)$ freedom) is $\hat{a}_i = J'_i/|\vec{J}'|$ (or $K'_i/|\vec{K}'|$ if $\vec{J}' = 0$), and the axial vector field on Δ is

$$\phi^A = \hat{a}_i \tilde{\phi}_i^A, \quad (50)$$

an explicit vector field at the collocation points of the *original* chart. Its two zeros are the rotation poles. For reporting the axis as a spatial direction, the poles can be located on the surface and the corresponding embedding points x^i (or the vector between them) quoted; for small $|\vec{K}|$ the axis is simply $\vec{J}/|\vec{J}|$ in the χ -frame.

3.10 Optional: explicit good coordinates

The algorithm never needs them, but conformally spherical coordinates can be read off from the eigenfunctions if desired (e.g. for visualization or for comparing with (6)):

$$\theta_1 = \arccos \chi_3, \quad \varphi_1 = \text{atan2}(\chi_2, \chi_1). \quad (51)$$

In these coordinates $q = F(d\theta_1^2 + \sin^2\theta_1 d\varphi_1^2)$ with $F = (\mathcal{A}/4\pi)e^{-2u_\infty}$ expressed as a function of (θ_1, φ_1) . Verifying this (by interpolating F and q onto a grid in (θ_1, φ_1) and checking the off-diagonal and trace parts) is a strong end-to-end test, but as a computational route it is inferior: the map (51) clusters and dilutes collocation points, and re-interpolation costs accuracy for no benefit.

4 Numerical implementation

4.1 Spin-weighted spherical harmonic representation

All fields live on a spectral grid on the coordinate sphere (equiangular or Gauss–Legendre in θ , equispaced in φ), with band limit ℓ_{max} , as provided by the packages `FastSphericalHarmonics` and

AbstractSphericalHarmonics. Tensor components in the (θ, φ) chart are singular at the poles; the cure is the $\bar{\partial}$ -formalism [4]: fix the unit-sphere dyad

$$m^A = \frac{1}{\sqrt{2}} \left(1, \frac{i}{\sin \theta}\right), \quad \hat{q}_{AB} m^A \bar{m}^B = 1, \quad m^A m_A = 0, \quad (52)$$

and store any tensor through its spin-weighted scalar components, e.g. for the metric

$$q_0 = q_{AB} m^A \bar{m}^B \text{ (spin 0, real)}, \quad q_2 = q_{AB} m^A m^B \text{ (spin 2)}, \quad (53)$$

and for one-forms $\omega_1 = \omega_A m^A$ (spin 1). These are globally smooth functions expandable in spin-weighted harmonics ${}_s Y_{\ell m}$. Covariant derivatives $\hat{\nabla}_A$ with respect to the *unit coordinate sphere* act as $\bar{\partial}/\bar{\partial}$ on spin components and are diagonal in the ${}_s Y_{\ell m}$ basis; derivatives with respect to q (or $\bar{q}$, $\dot{q}$) are then assembled algebraically with the tensor C_{BC}^A of (29), exactly as in (30)–(31). (The pre-1.0 prototype of this package implemented (30) this way.)

Scalars such as h , u , g , χ_i , R , and the integrands of (48) are ordinary spin-0 fields. Surface integrals are evaluated by quadrature as $\oint f \epsilon = \int f \frac{\sqrt{\det q}}{\sin \theta} \sin \theta d\theta d\varphi$, where the ratio $\sqrt{\det q}/\sin \theta$ is a smooth strictly positive scalar; with a spectral transform this is just $\sqrt{4\pi}$ times the $\ell = 0$ coefficient of $f\sqrt{\det q}/\sin \theta$.

Dealiasing. The pipeline is full of pointwise products and of the nonlinearity e^{2u} ; use a grid with $\gtrsim 2\ell_{\max}$ points per dimension (3/2-rule at minimum) and verify spectral tails. All quantities derived from h and the metric are smooth, so coefficients decay exponentially; the resolution requirement is set by the distortion of the horizon (the dynamic range of F).

4.2 Elliptic solves and the eigenproblem

The operators Δ_q (for (26)), the Newton operator $\bar{\Delta} + 2e^{2u}$ (for (36)), and $\bar{\Delta} + 2e^{2u_\infty}$ (shift-invert in (39)) are all of the form “unit-sphere Laplacian plus smooth corrections”. Apply them matrix-free on spherical-harmonic coefficients (transform $\rightarrow$ pointwise algebra $\rightarrow$ transform) and solve with preconditioned CG/MINRES, using the diagonal unit-sphere Laplacian $-\ell(\ell+1)$ (shifted) as preconditioner. For moderate resolution ($\ell_{\max} \lesssim 50$, i.e. $(\ell_{\max} + 1)^2 \lesssim 2600$ modes) dense assembly and direct factorization/eigensolution is perfectly affordable and simplest.

For (39), request the three eigenpairs nearest -2 of the pencil $(\bar{\Delta}, e^{2u_\infty})$, e.g. with shift-invert Lanczos/Arnoldi (**Arpack.jl**, as in the pre-1.0 prototype) in the inner product $\langle \cdot, \cdot \rangle$, or equivalently the eigenpairs of the symmetrized operator $e^{-u_\infty} \bar{\Delta} e^{-u_\infty}$. The constant function (eigenvalue 0) and the $\ell = 2$ -like cluster (near -6) are well separated, so convergence is fast and unambiguous.

4.3 Diagnostics

Cheap, sharp internal checks, in pipeline order:

1. spectral tail of h , q_0 , q_2 (input resolution);
2. Gauss–Bonnet: $\oint R[q] \epsilon = 8\pi$;
3. Ricci flow / Newton: $\|R[\dot{q}] - 2\|_\infty$, area drift;
4. eigencluster: $|\mu_i + 2|$ for the three eigenvalues, residuals $\|\bar{\Delta}\chi_i + 2e^{2u_\infty}\chi_i\|$;
5. rigidity: $\|\delta^{ij}\chi_i\chi_j - 1\|_\infty$ and $\|\delta^{ij}\partial_A\chi_i\partial_B\chi_j - \dot{q}_{AB}\|_\infty$ (43);

6. Killing check: $\|\mathcal{L}_{\phi_i}\dot{q}\|$, conformal check: $\mathcal{L}_{\xi_i}\dot{q} = 2\chi_i\dot{q}$ (from $\dot{D}_A\dot{D}_B\chi_i = -\chi_i\dot{q}_{AB}$);
7. Hodge: $\oint D^A\omega_A^{\text{inv}}$ -residual; same g from q - and $\dot{q}$ -divergences;
8. invariance: $|\vec{J}'|$ from the boosted frame vs. (15); $\vec{J}' \times \vec{K}' \approx 0$;
9. convergence of $J, \vec{J}, \vec{K}$ under ℓ_{max} doubling (should be exponential).

5 Algorithm summary

1. **Geometry.** From h and γ_{ij} : tangents e_A^i , metric q_{AB} , area form ϵ , area $\mathcal{A}$, outward normal s^i (19)–(22).
2. **Rotation form.** $\omega_A = -K_{ij}e_A^i s^j$ (25).
3. **Gauge fix.** Solve $\Delta_q g = D^A\omega_A$; set $\omega^{\text{inv}} = \omega - dg$ (26).
4. **Normalize.** $\bar{q} = (4\pi/\mathcal{A})q$; compute $\bar{R}$ (30).
5. **Uniformize.** Ricci flow (34) (optionally Newton polish (36)) $\rightarrow u_\infty$, $\dot{q} = e^{2u_\infty}\bar{q}$ with $R[\dot{q}] = 2$.
6. **Eigenfunctions.** Solve $\bar{\Delta}\chi = -2e^{2u_\infty}\chi$ for the triple near -2 ; orthonormalize (41); fix handedness (44).
7. **Generators.** $\phi_i^A = \epsilon^{AB}\partial_B\chi_i$, $\xi_i^A = -\dot{q}^{AB}\partial_B\chi_i$ (45).
8. **Invariants.** J_i, K_i by quadrature (48) with the *physical* ϵ ; then A, B (12) and J (15).
9. **Axis and axial field.** If $A^2 + B^2 > 0$: boost (13), (11), (9); output J , axis $\hat{a}_i = J'_i/|\vec{J}'|$, and $\phi^A = \hat{a}_i\tilde{\phi}_i^A$ (14).

6 Tests

1. **Round sphere in flat space / Schwarzschild.** $\omega = 0$ identically; every stage should return its trivial value (u_∞ constant = 0, χ_i the $\ell = 1$ harmonics, $\vec{J} = \vec{K} = 0$, $J = 0$) to round-off.
2. **Coordinate-distorted Schwarzschild.** Apply a smooth angle-dependent radial coordinate distortion (so that h is nontrivial but the geometry is unchanged): again $J = 0$, and $\dot{q}, \chi_i$ must reproduce the distortion map. This tests uniformization + eigenfunctions in isolation from ω .
3. **Kerr, standard slicing.** For a Kerr–Schild (or Boyer–Lindquist) slice of Kerr with parameters (M, a) , the horizon $r = r_+$ gives $\vec{K} = 0$ (after gauge fixing), $\vec{J}$ along the spin axis, and $J = Ma$ to spectral accuracy; the axial field ϕ must agree with the axial Killing vector. The `SpacetimeMetrics` package provides the analytic data (`KerrSchild` with `adm_decompose` and `ExtrinsicCurvature`). Verify $J_3 = +Ma$ (not $-Ma$) to pin down orientation conventions.
4. **Rotated/translated Kerr.** Rotate the spin axis and offset the centre c^i : J invariant, $\vec{J}$ rotates as a vector, the recovered axis follows.
5. **Tilted slicing of Kerr.** Change the slicing through the horizon (tilted/waved leaves). Without the Hodge projection the raw $\vec{J}$ shifts; with it, $J = Ma$ is recovered — this is the paper’s own acid test for the gauge fixing [1].

6. **Distorted non-axisymmetric data.** E.g. Bowen–York or superposed-Kerr initial data: check exponential convergence of J with $\ell_{\max}$, the internal diagnostics of Section 4, and stability of the axis under resolution changes.

A Corrections and clarifications to the GOAL.md skeleton

1. **“Use Ricci flow to find a conformally round metric.”** In two dimensions the (normalized) Ricci flow moves only the conformal factor, so it does not *find* a conformally round metric — q is already conformally round by uniformization. What the flow finds is the *round representative* $\hat{q} = e^{2u_\infty} \bar{q}$ of the conformal class, in the original chart. Operationally this is what the skeleton intended, but it reduces the PDE problem from a tensor flow to the single scalar equation (34).
2. **“Undo the Ricci flow to find $q(\theta_1, \varphi_1)$.”** Unnecessary, and best avoided. The physical metric is known all along in the original chart; the invariants need only ω^{inv} , ϵ , and the generators ϕ_i, ξ_i , and Lemma 1 provides the generators directly in the original chart. No quantity is ever transformed to (θ_1, φ_1) , and no re-interpolation occurs. The explicit coordinates (51) survive only as an optional diagnostic.
3. **Determination of x, y, z (here $\chi_{1,2,3}$).** The skeleton’s conditions (range $[-1, 1]$, zero mean, orthogonality, normalization) do not determine the functions and cannot be “solved as a linear system”: infinitely many triples satisfy them. The missing condition is that they span the *lowest nonzero eigenspace of $\hat{\Delta}$* (eigenvalue -2), eq. (39) — so the correct guess in the skeleton is “eigenvalue problem”. Zero mean is then automatic; orthogonality and normalization fix the Gram matrix (41); the range $[-1, 1]$ and indeed $\sum_i \chi_i^2 = 1$ follow by rigidity (43) rather than being imposable conditions.
4. **Integration measure.** The orthogonality/normalization integrals must use the *round* area form $\hat{\epsilon}$ (equivalently weight e^{2u_∞} in $\bar{q}$ -integrals), with $\oint \chi_i \chi_j \hat{\epsilon} = \frac{4\pi}{3} \delta_{ij}$, not the coordinate or physical measure.
5. **Area normalization.** The metric must be scaled to area 4π before the flow/eigenproblem; otherwise the target curvature is not 2, the eigenvalue is not -2 , and the generators (45) acquire wrong normalizations (orbits would not have period 2π). The physical area form still appears in the final integrals (48).
6. **North pole and zero meridian.** No such choice is needed: the residual freedom is a global $SO(3)$ under which all outputs are covariant. What *is* needed instead is a handedness fix (44), which the skeleton omits; with the wrong orientation, $\vec{J} \rightarrow -\vec{J}$.
7. **Gauge fixing of ω .** The skeleton omits the Hodge projection (17) entirely. It is essential: without it the J_i depend on the choice of null normals / slicing through the horizon (Section 3.4), and e.g. a tilted slice of Kerr gives a wrong spin.
8. **K_i and the boost.** The skeleton computes only the J_i -side implicitly. The K_i are needed both for the invariant value of the spin (15) and for the axial vector field, which is defined in the boosted frame where $\vec{J} \parallel \vec{K}$ (13); only for $\vec{K} \times \vec{J} = 0$ (e.g. exact axisymmetry) can the boost be skipped.
9. **Inputs.** Derivatives of γ_{ij} and K_{ij} are not required (Section 3.1); listing them among the inputs is harmless but unnecessary.

B Two-dimensional identities used

For $\tilde{q} = e^{2u}q$ on a two-manifold: $\tilde{q}^{AB} = e^{-2u}q^{AB}$; $\tilde{\epsilon}_{AB} = e^{2u}\epsilon_{AB}$, $\tilde{\epsilon}^{AB} = e^{-2u}\epsilon^{AB}$; $\tilde{\epsilon}_A{}^B = \epsilon_A{}^B$ (conformal invariance of the Hodge star on one-forms); $\Delta_{\tilde{q}}f = e^{-2u}\Delta_q f$; $R[\tilde{q}] = e^{-2u}(R[q] - 2\Delta_q u)$. Levi-Civita identities: $\epsilon_A{}^C \epsilon_C{}^B = -\delta_A^B$, $\epsilon^{AC} \epsilon_{BC} = \delta_B^A$, $\epsilon^{AB} D_A D_B f = 0$. Hodge: $\star\star = -1$ on one-forms; for $\omega = \star df + dg$: $D^A \omega_A = \Delta g$, $\epsilon^{AB} D_A \omega_B = -\Delta f$. Gauss–Bonnet: $\oint R \epsilon = 8\pi$ on S^2 . First-eigenfunction (Obata-type) identity on the unit round sphere: $\mathring{D}_A \mathring{D}_B \chi_i = -\chi_i \mathring{q}_{AB}$, whence $\mathring{\Delta} \chi_i = -2\chi_i$ and, for $\xi_i{}^A = -\mathring{q}^{AB} \partial_B \chi_i$,

$$\mathcal{L}_{\xi_i} \mathring{q}_{AB} = -2 \mathring{D}_A \mathring{D}_B \chi_i = 2\chi_i \mathring{q}_{AB}, \quad \mathring{D}_A \xi_i{}^A = 2\chi_i,$$

the conformal Killing equation with divergence $2\chi_i$.

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